



Abhiraj Kumar

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B.Tech - Information Technology

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Information Technology, Una	8.71 (Current)	2023-2027
Senior Secondary	CBSE Board	93.4%	2022
Secondary	CBSE Board	97.6%	2020

PERSONAL PROJECTS

• VeriFi: Fake News Detection using ML

GitHub Repo

A machine learning model to classify news articles as real or fake using Logistic Regression.

- Tech Stack: Python, scikit-learn, Pandas, NumPy, Matplotlib
- Performed text preprocessing including tokenization, stopword removal, and lemmatization.
- Used TF-IDF vectorization for feature extraction and trained a Logistic Regression model for binary classification.
- Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix.

• NeuroSight: 3D Brain Tumor Analysis (Flask + TensorFlow)

GitHub Repo

A web-based system for 3D brain tumor segmentation and tumor grade classification (HGG/LGG) from MRI volumes.

- Tech Stack: Python, TensorFlow/Keras, NumPy, Matplotlib, Flask, SQLAlchemy, ReportLab
- Developed a Flask web application for end-to-end inference on preprocessed 3D MRI volumes.
- Trained and evaluated segmentation and classification models using the BraTS 2020 dataset for HGG/LGG prediction.
- Implemented modality-occlusion based explainability by measuring IoU drop across MRI modalities.
- Generated visual collages, tumor volume metrics, progress graphs, and downloadable PDF diagnostic reports.
- Stored and managed patient history in SQLite with fast retrieval using a 3-digit User ID search.

• UnitySplit: Smart Expense Splitting API (MERN Stack)

GitHub Repo

A production-ready REST API for group expense management with automated debt minimization.

- Tech Stack: Node.js, Express 5, MongoDB, Mongoose, JWT, React, Vite
- Implemented the Min Cash Flow algorithm to compute the minimum number of transactions needed to settle all group debts.
- Designed a flexible expense splitting engine supporting EQUAL, EXACT amount, and PERCENTAGE-based splits.
- Built JWT-authenticated REST API with role-based authorization, invite-code based group joining, and soft-delete.
- Secured the API with Helmet security headers, per-route rate limiting, body size caps, and a centralized error handler.
- Added a real-world settlement tracking system and a 23-assertion end-to-end smoke test suite.

TECHNICAL SKILLS

- **Languages:** C, C++, HTML, CSS, JavaScript, Python
- **Developer Tools:** VS Code, GitHub
- **Frameworks:** Express.js, Node.js
- **Cloud/Databases:** MySQL, MongoDB
- **Soft Skills:** Problem-Solving, Communication, Collaboration, Time Management, Critical Thinking

KEY COURSES TAKEN

- **Information Technology:** DSA, DBMS, Web Development, Operating Systems, Computer Networks

ACHIEVEMENTS

- **Competitive Programming,** Solved 800+ DSA questions across Platforms. Coding Profile
- **JEE Main,** 97.32 Percentile in JEE Main 2023
- **10th Boards,** 97.6 Percent in 10th Boards 2020
- **12th Boards,** 93.4 Percent in 12th Boards 2022
- **AlgoUniversity Graph Camp,** Completed AlgoUniversity Graph Theory Programming Camp. 2026